

1.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections". The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and a user profile "Vidhushini T S".

On the left, the "QUESTIONS" panel shows "4/8 completed" and a dropdown menu set to "Logger". Below it, the "LOGGER" section contains a description: "You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called 'Logger' with methods for logging different types of messages, such as 'debug', 'info', 'warning', and 'error'. Implement the interface in a class called 'Log4jLogger' that uses the Log4j logging framework."

The "PROGRAM EDITOR" panel shows the following Java code:

```
1 import java.util.Scanner;
2 interface Logger{
3     void debug(String message);
4     void info(String message);
5     void warning(String message);
6     void error(String message);
7 }
8 public class Log4jLogger implements Logger{
9     @Override
10    public void debug(String message){
11        System.out.println("Debug: " + message);
12    }
13    @Override
14    public void info(String message){
15        System.out.println("Info: " + message);
16    }
17    @Override
18    public void warning(String message){
19        System.out.println("Warning: " + message);
20    }
21    @Override
```

The "OUTPUT" panel shows the following text:

```
Debug: Starting application
Info: User Login
Warning: Low memory space
Error: Database connection failed
```

The "INPUT" panel is empty.

The bottom status bar shows a temperature of 32°C, "Mostly cloudy", and the date "30-07-2026".

2.

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On the left, the "QUESTIONS" panel shows "4/8 completed" and a dropdown menu set to "Character". Below it, the "CHARACTER" section contains a description: "You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called 'Character' with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of character, such as 'Hero', 'Villain', and 'Monster'."

The "PROGRAM EDITOR" panel shows the following Java code:

```
1 import java.util.Scanner;
2 import java.util.ArrayList;
3 import java.util.Collections;
4 import java.util.List;
5 public class Main{
6     public static void main(String[] args){
7         Scanner sc=new Scanner(System.in);
8         List<Integer> list=new ArrayList<>();
9         while (sc.hasNextInt()){
10             list.add(sc.nextInt());
11         }
12         Collections.sort(list);
13         for(int i=0;i<list.size();i++){
14             System.out.print(list.get(i));
15             if(i<list.size()-1){
16                 System.out.print(" ");
17             }
18         }
19         sc.close();
20     }
21 }
```

The "OUTPUT" panel shows the following text:

```
0 1 2 3 5
```

The "INPUT" panel shows the text "1 2 5 3 0".

The bottom status bar shows a temperature of 32°C, "Mostly cloudy", and the date "30-07-2026".

3.

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar has a 'QUESTIONS' section with 'Sort' selected. The main area is the 'PROGRAM EDITOR' for Java. The code is as follows:

```

1 import java.util.Scanner;
2 import java.util.ArrayList;
3 import java.util.List;
4 public class Main{
5     public static void main(String[] args){
6         Scanner sc=new Scanner(System.in);
7         String input=sc.nextLine();
8         List<Character>charList=new ArrayList<>();
9         for(char c:input.toCharArray()){
10             charList.add(c);
11         }
12         for(int i=0;i<charList.size();i++){
13             System.out.print(charList.get(i));
14             if(i<charList.size()-1)
15                 System.out.print(" ");
16         }
17         System.out.println();
18         sc.close();
19     }
20 }

```

The 'INPUT' field contains 'hello' and the 'OUTPUT' field shows 'h e l l o'.

4.

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar has a 'QUESTIONS' section with 'HashSet' selected. The main area is the 'PROGRAM EDITOR' for Java. The code is as follows:

```

1 import java.util.Scanner;
2 import java.util.HashSet;
3 public class Main{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6         HashSet<String> set=new HashSet<>();
7         int n=sc.nextInt();
8         sc.nextLine();
9         for(int i=0;i<n;i++){
10             set.add(sc.nextLine());
11         }
12         String toRemove=sc.nextLine();
13         set.remove(toRemove);
14         System.out.println(set);
15         sc.close();
16     }
17 }

```

The 'INPUT' field contains '4', 'Apple', 'Banana', 'Cherry', 'Date', and 'Banana'. The 'OUTPUT' field shows a 'Security Error: Disallowed code pattern detected.'

5.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections". The page header includes the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" with a "JAVA PROGRAMMING" tag. A user profile "Vidhushini T S" is visible in the top right.

On the left, the "QUESTIONS" panel shows "8 / 8 completed" and a dropdown menu with "TreeMap" selected. Below it, the "TREETMAP" section contains the instruction: "Write a Java program to create a TreeMap of strings and integers and iterate over its entries."

The "PROGRAM EDITOR" panel shows a Java code snippet:

```
1 import java.util.Scanner;
2 import java.util.TreeMap;
3 import java.util.Map;
4 public class Main{
5     public static void main(String[] args){
6         Scanner sc=new Scanner(System.in);
7         TreeMap<String,Integer> map=new TreeMap<>();
8         int n=sc.nextInt();
9         for(int i=0;i<n;i++){
10             String key=sc.next();
11             int value=sc.nextInt();
12             map.put(key,value);
13         }
14         for(Map.Entry<String,Integer> entry:map.entrySet()){
15             System.out.println(entry.getKey()+" "+entry.getValue());
16         }
17         sc.close();
18     }
19 }
```

Below the code editor is the "OUTPUT" panel, which displays the result of the program execution:

```
Apple 50
Banana 20
Cherry 30
```

On the right, the "INPUT" panel shows the input provided to the program:

```
3
Apple 50
Banana 20
Cherry 30
```

The bottom of the interface features a status bar with a copyright notice "© 2026 SIMATS Online Lab. All rights reserved." and a Windows taskbar showing the date "30-07-2026" and time "11:52".

6.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections". The page header includes the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" with a "JAVA PROGRAMMING" tag. A user profile "Vidhushini T S" is visible in the top right.

On the left, the "QUESTIONS" panel shows "8 / 8 completed" and a dropdown menu with "PriorityQueue" selected. Below it, the "PRIORITYQUEUE" section contains the instruction: "Write a Java program to create a PriorityQueue of integers and add elements to it."

The "PROGRAM EDITOR" panel shows a Java code snippet:

```
1 import java.util.PriorityQueue;
2 import java.util.Scanner;
3 public class Main{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6         PriorityQueue<Integer> pq=new PriorityQueue<>();
7         int n=sc.nextInt();
8         for(int i=0;i<n;i++){
9             pq.add(sc.nextInt());
10         }
11         while(!pq.isEmpty()){
12             System.out.print(pq.poll()+" ");
13         }
14         sc.close();
15     }
16 }
```

Below the code editor is the "OUTPUT" panel, which displays the result of the program execution:

```
1 5 10 15 20
```

On the right, the "INPUT" panel shows the input provided to the program:

```
5
10 5 20 1 15
```

The bottom of the interface features a status bar with a copyright notice "© 2026 SIMATS Online Lab. All rights reserved." and a Windows taskbar showing the date "30-07-2026" and time "11:53".

7.

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QUESTIONS 8 / 8 completed

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

```
1 import java.util.LinkedHashMap;
2 import java.util.Scanner;
3 public class Main{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6         LinkedHashMap<String,Integer>map=new LinkedHashMap<>();
7         int n=sc.nextInt();
8         for(int i=0;i<n;i++){
9             String key=sc.next();
10            int value=sc.nextInt();
11            map.put(key,value);
12        }
13        for(String key:map.keySet()){
14            System.out.println(key+" "+map.get(key));
15        }
16        sc.close();
17    }
18 }
```

OUTPUT

```
Banana 20
Apple 50
Cherry 30
```

INPUT

```
3
Banana 20
Apple 50
Cherry 30
```

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8.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=collections". The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user "Vidhushini T S".

QUESTIONS 8 / 8 completed

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

```
1 import java.util.HashSet;
2 import java.util.Scanner;
3 import java.util.Collections;
4 public class Main{
5     public static void main(String[] args){
6         Scanner sc=new Scanner(System.in);
7         HashSet<Integer> set=new HashSet<>();
8         int n=sc.nextInt();
9         for(int i=0;i<n;i++){
10            set.add(sc.nextInt());
11        }
12        System.out.println("Max: "+Collections.max(set));
13        System.out.println("Min: "+Collections.min(set));
14    }
15 }
```

OUTPUT

```
Max: 40
Min: 5
```

INPUT

```
5
10 25 5 40 15
```

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